

IF2240 Basdat

# **Formal Relational Query Language: Null Values, Multiset Algebra, Relational Calculus**

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## Null Values

- it is possible for tuples to have a null value, denoted by *null*, for some of their attributes
- null signifies an unknown value or a value that doesn't exist
- the result of any arithmetic expression involving null is null
- aggregate functions simply ignore null values (as in SQL)
- for duplicate elimination and grouping, null is treated like any other value, and two nulls are assumed to be the same (as in SQL)
- comparisons with null values return the special truth value: unknown
  - if false was used instead of unknown, then  $\text{not}(A < 5)$  would not be equivalent to  $A \geq 5$
- Three-valued logic using the truth value unknown:
  - OR:  $(\text{unknown OR true}) = \text{true}$ ,  
 $(\text{unknown OR false}) = \text{unknown}$ ,  
 $(\text{unknown OR unknown}) = \text{unknown}$
  - AND:  $(\text{true AND unknown}) = \text{unknown}$ ,  
 $(\text{false AND unknown}) = \text{false}$ ,  
 $(\text{unknown AND unknown}) = \text{unknown}$
  - NOT:  $(\text{NOT unknown}) = \text{unknown}$
  - In SQL "P is unknown" evaluates to true if predicate P evaluates to unknown
  - Result of select predicate is treated as false if it evaluates to unknown

## Multiset Relational Algebra

- Pure relational algebra removes all duplicates, e.g. after projection
- Multiset relational algebra retains duplicates, to match SQL semantics
  - SQL duplicate retention was initially for efficiency, but is now a feature
- Multiset relational algebra defined as follows
  - selection: has as many duplicates of a tuple as in the input, if the tuple satisfies the selection
  - projection: one tuple per input tuple, even if it is a duplicate
  - cross product: if there are  $m$  copies of  $t_1$  in  $r$ , and  $n$  copies of  $t_2$  in  $s$ , there are  $m \times n$  copies of  $t_1.t_2$  in  $r \times s$
  - Other operators similarly defined, such as union:  $m + n$  copies, intersection:  $\min(m, n)$  copies, difference:  $\min(0, m - n)$  copies

# Relational Calculus

## Tuple Relational Calculus

- A nonprocedural query language, where each query is of the form:  
 $\{t \mid P(t)\}$
- It is the set of all tuples  $t$  such that predicate  $P$  is true for  $t$
- $t$  is a tuple variable,  $t[A]$  denotes the value of tuple  $t$  on attribute  $A$
- $t \in r$  denotes that tuple  $t$  is in relation  $r$
- $P$  is a formula similar to that of the predicate calculus

## Predicate Calculus Formula

1. Set of attributes and constants
2. Set of comparison operators, e.g.  $<$ ,  $\leq$ ,  $=$ ,  $\neq$ ,  $>$ ,  $\geq$
3. Set of connectives: and, or, not
4. implication ( $\Rightarrow$ ):  $x \Rightarrow y$ , if  $x$  is true then  $y$  is true  
$$x \Rightarrow y \equiv \neg x \vee y$$
5. Set of quantifiers:
  - a.  $\exists t \in r(Q(t)) \equiv$  "there exists" a tuple  $t$  in relation  $r$  such that predicate  $Q(t)$  is true
  - b.  $\forall t \in r(Q(t)) \equiv Q$  is true "for all" tuples  $t$  in relation  $r$

Find the *ID*, *name*, *dept\_name*, *salary* for instructors whose salary is greater than \$80,000

```
{t | t ∈ instructor ∧ t[salary] > 80000}
```

As in the previous query, but output only the *ID* attribute value

```
{t | ∃ s ∈ instructor (t[ID] = s[ID] ∧ s[salary] > 80000)}
```

Notice that a relation on schema (*ID*) is implicitly defined by the query

Find the names of all instructors whose department is in the Watson building

```
{t | ∃ s ∈ instructor (t[name] = s[name]  
  ∧ ∃ u ∈ department (u[dept_name] = s[dept_name]  
    ∧ u[building] = "Watson" ))}
```

Find the set of all courses taught in the Fall 2017 semester, or in the Spring 2018 semester, or both

```
{t | ∃ s ∈ section (t[course_id] = s[course_id] ∧  
  s[semester] = "Fall" ∧ s[year] = 2017)  
  ∨ ∃ u ∈ section (t[course_id] = u[course_id] ∧  
    u[semester] = "Spring" ∧ u[year] = 2018)}
```

Find the set of all courses taught in the Fall 2017 semester, and in the Spring 2018 semester

```
{t | ∃ s ∈ section (t [course_id] = s [course_id] ∧
    s [semester] = "Fall" ∧ s [year] = 2017)
    ∧ ∃ u ∈ section (t [course_id] = u [course_id] ∧
    u [semester] = "Spring" ∧ u [year] = 2018))
```

Find the set of all courses taught in the Fall 2017 semester, but not in the Spring 2018 semester

```
{t | ∃ s ∈ section (t [course_id] = s [course_id] ∧
    s [semester] = "Fall" ∧ s [year] = 2017)
    ∧ ¬ ∃ u ∈ section (t [course_id] = u [course_id] ∧
    u [semester] = "Spring" ∧ u [year] = 2018))
```

## Safety of Expressions

- It is possible to write tuple calculus expressions that generate infinite relations
- For example,  $\{t \mid \neg t \in r\}$  results in an infinite relation if the domain of any attribute of relation  $r$  is infinite
- To guard against the problem, we restrict the set of allowable expressions to safe expressions
- An expression  $\{t \mid P(t)\}$  in the tuple relational calculus is safe if every component of  $t$  appears in one of the relations, tuples, or constants that appear in  $P$ 
  - this is more than just a syntax condition, e.g.  $\{t \mid t[A] = 5 \vee \text{true}\}$  is not safe – it defines an infinite set with attribute values that do not appear in any relation or tuples or constants in  $P$

## Universal Quantification

Find all students who have taken all courses offered in the Biology department

```
{t | ∃ r ∈ student (t [ID] = r [ID]) ∧
    (∀ u ∈ course (u [dept_name] = "Biology" ⇒
        ∃ s ∈ takes (t [ID] = s [ID] ∧
            s [course_id] = u [course_id]))))
```

- Note that without the existential quantification on student, the above query would be unsafe if the Biology department has not offered any courses.

## Domain Relational Calculus

- A nonprocedural query language equivalent in power to the tuple relational calculus.
- Each query is an expression of the form:

$$\{ \langle x_1, x_2, \dots, x_n \rangle \mid P(x_1, x_2, \dots, x_n) \}$$

- $x_1, x_2, \dots, x_n$  represent domain variables

- P represents a formula similar to that of the predicate calculus
- The expression is safe if all of the following hold:
  - All values that appear in tuples of the expression are values from  $\text{dom}(P)$  (that is, the values appear either in P or in a tuple of a relation mentioned in P)
  - For every "there exists" subformula of the form  $\exists x(P_1(x))$ , the subformula is true if and only if there is a value of x in  $\text{dom}(P_1)$  such that  $P_1(x)$  is true.
  - For every "for all" subformula of the form  $\forall x(P_1(x))$ , the subformula is true if and only if  $P_1(x)$  is true for all values x from  $\text{dom}(P_1)$ .

Find the *ID, name, dept\_name, salary* for instructors whose salary is greater than \$80,000

$\{ \langle i, n, d, s \rangle \mid \langle i, n, d, s \rangle \in \text{instructor} \wedge s > 80000 \}$

As in the previous query, but output only the *ID* attribute value

$\{ \langle i \rangle \mid \exists n, d, s (\langle i, n, d, s \rangle \in \text{instructor} \wedge s > 80000) \}$

Find the names of all instructors whose department is in the Watson building

$\{ \langle n \rangle \mid \exists i, d, s (\langle i, n, d, s \rangle \in \text{instructor} \wedge \exists b, a (\langle d, b, a \rangle \in \text{department} \wedge b = \text{"Watson"})) \}$

i, n, d, s (represents the needed attributes) adalah elemen dari instructor dan s lebih dari 80000.

> yang perlu diberi quantifier adalah variabel-variabel yang tidak muncul sbg hasil

Find the set of all courses taught in the Fall 2017 semester, or in the Spring 2018 semester, or both

$\{ \langle c \rangle \mid \exists a, s, y, b, r, t (\langle c, a, s, y, b, r, t \rangle \in \text{section} \wedge s = \text{"Fall"} \wedge y = 2017) \vee \exists a, s, y, b, r, t (\langle c, a, s, y, b, r, t \rangle \in \text{section} \wedge s = \text{"Spring"} \wedge y = 2018) \}$

This case can also be written as

$\{ \langle c \rangle \mid \exists a, s, y, b, r, t (\langle c, a, s, y, b, r, t \rangle \in \text{section} \wedge ((s = \text{"Fall"} \wedge y = 2017) \vee (s = \text{"Spring"} \wedge y = 2018))) \}$

Find the set of all courses taught in the Fall 2017 semester, and in the Spring 2018 semester

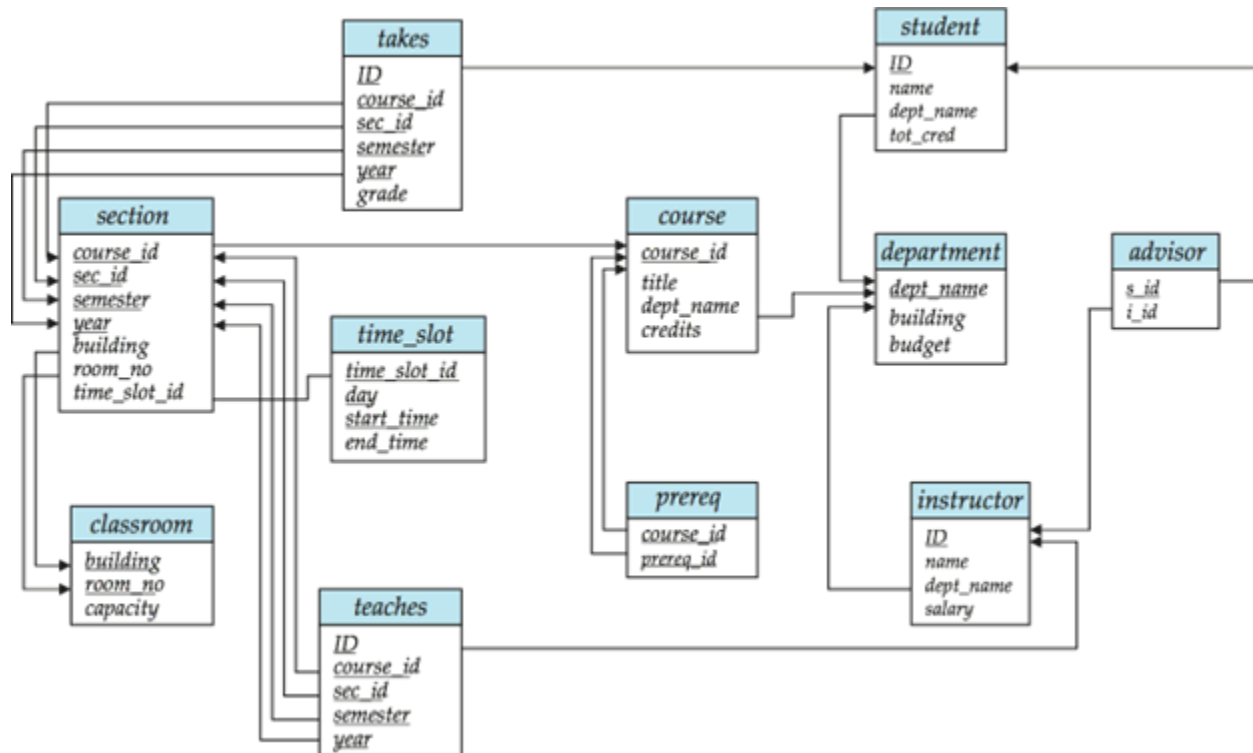
$\{ \langle c \rangle \mid \exists a, s, y, b, r, t (\langle c, a, s, y, b, r, t \rangle \in \text{section} \wedge s = \text{"Fall"} \wedge y = 2017) \wedge \exists a, s, y, b, r, t (\langle c, a, s, y, b, r, t \rangle \in \text{section} \wedge s = \text{"Spring"} \wedge y = 2018) \}$

Find all students who have taken all courses offered in the Biology department

$\{ \langle i \rangle \mid \exists n, d, tc (\langle i, n, d, tc \rangle \in \text{student} \wedge (\forall ci, ti, dn, cr (\langle ci, ti, dn, cr \rangle \in \text{course} \wedge dn = \text{"Biology"} \Rightarrow \exists si, se, y, g (\langle i, ci, si, se, y, g \rangle \in \text{takes}))) \}$



## Latihan



Berdasarkan skema tersebut, tuliskan ekspresi a) kalkulus relasional tuple dan b) kalkulus relasional domain untuk mendapatkan informasi berikut ini:

1. Tampilkan seluruh mahasiswa yang berasal dari Departemen "IF" atau "STI" dengan total credit (sks lulus) sekurang-kurangnya 130 sks.

Jawab:

- a)  $\{t \mid t \in \text{student} \wedge (t[\text{dept\_name}] = \text{"IF"} \vee t[\text{dept\_name}] = \text{"STI"}) \wedge t[\text{tot\_cred}] \geq 130\}$
- b)  $\{ \langle i, n, d, tc \rangle \mid \langle i, n, d, tc \rangle \in \text{student} \wedge (d = \text{"IF"} \vee d = \text{"STI"}) \wedge tc \geq 130 \}$

2. Tampilkan nama dari semua mata kuliah yang diselenggarakan pada Semester 1-2020.

Jawab:

- a)  $\{t \mid \exists s \in \text{section} (s[\text{semester}] = \text{"1"} \wedge s[\text{year}] = 2020 \wedge \exists c \in \text{course} (c[\text{course\_id}] = s[\text{course\_id}] \wedge t[\text{title}] = c[\text{title}]))\}$
- b)  $\{ \langle ti \rangle \mid \exists ci, dn, cr (\langle ci, ti, dn, cr \rangle \in \text{course} \wedge \exists si, sm, y, b, r, ts (\langle ci, si, sm, y, b, r, ts \rangle \in \text{section} \wedge sm = \text{"1"} \wedge y = 2020)) \}$

3. Tampilkan nama dari semua orang yang terdapat di perguruan tinggi.

Jawab:

- a)  $\{t \mid (\exists s \in \text{student} (t[\text{name}] = s[\text{name}])) \vee (\exists i \in \text{instructor} (t[\text{name}] = i[\text{name}]))\}$
- b)  $\{ \langle n \rangle \mid (\exists i, d, tc (\langle i, n, d, tc \rangle \in \text{student})) \vee (\exists i, d, s (\langle i, n, d, s \rangle \in \text{instructor})) \}$

4. ID, nama, dan departemen asal mahasiswa yang pernah mengambil mata kuliah yang bukan berasal dari departemennya, berikut informasi nama, jumlah sks, dan departemen penyelenggara mata kuliah tersebut.

a)  $\{t \mid \exists st \in student \exists tk \in takes \exists c \in course (t[ID] = st[ID] \wedge t[s\_name] = st[name] \wedge t[s\_dept] = st[dept\_name] \wedge t[c\_title] = c[title] \wedge t[c\_credits] = c[credits] \wedge t[c\_dept] = c[dept\_name] \wedge st[ID] = tk[ID] \wedge tk[course\_id] = c[course\_id] \wedge st[dept\_name] \neq c[dept\_name])\}$

b)  $\{ \langle sid, sn, sd, ct, cc, cd \rangle \mid \exists tc (\langle sid, sn, sd, tc \rangle \in student) \wedge \exists cid, si, sm, y, g (\langle sid, cid, si, sm, y, g \rangle \in takes \wedge \langle cid, ct, cd, cc \rangle \in course \wedge sd \neq cd) \}$

Jawab:

5. Nama instruktur yang tidak mengajar pada semester 1 tahun 2020.

Jawab:

a)  $\{t \mid \exists i \in instructor (t[name] = i[name] \wedge \neg \exists th \in teaches (th[ID] = i[ID] \wedge th[semester] = "1" \wedge th[year] = 2020))\}$

b)  $\{ \langle n \rangle \mid \exists i, d, s (\langle i, n, d, s \rangle \in instructor \wedge \neg \exists ci, si, sm, y (\langle i, ci, si, sm, y \rangle \in teaches \wedge sm = "1" \wedge y = 2020)) \}$

6. Nama dari semua mata kuliah yang selalu diselenggarakan di setiap semester.

Jawab:

a)  $\{t \mid \exists c \in course (t[title] = c[title] \wedge \forall s1 \in section (\exists s2 \in section (s2[course\_id] = c[course\_id] \wedge s2[semester] = s1[semester])))\}$

b)  $\{ \langle ti \rangle \mid \exists ci, dn, cr (\langle ci, ti, dn, cr \rangle \in course \wedge \forall ci2, si2, sm2, y2, b2, r2, ts2 (\langle ci2, si2, sm2, y2, b2, r2, ts2 \rangle \in section \Rightarrow \exists si3, y3, b3, r3, ts3 (\langle ci, si3, sm2, y3, b3, r3, ts3 \rangle \in section))) \}$

7. Nama departemen dengan budget terkecil.

Jawab:

a)  $\{t \mid \exists d \in department (t[dept\_name] = d[dept\_name] \wedge \neg \exists d2 \in department (d2[budget] < d[budget]))\}$

b)  $\{ \langle dn \rangle \mid \exists b, bd (\langle dn, b, bd \rangle \in department \wedge \neg \exists dn2, b2, bd2 (\langle dn2, b2, bd2 \rangle \in department \wedge bd2 < bd)) \}$